CSA 1610

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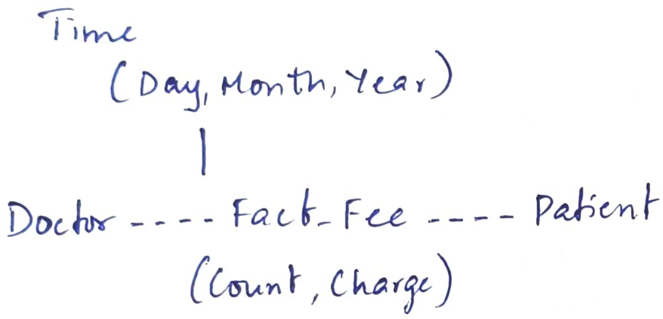
# Data Warehousing and Data Mining

CO1 AT1

1, a, Three popular data warehouse schemas

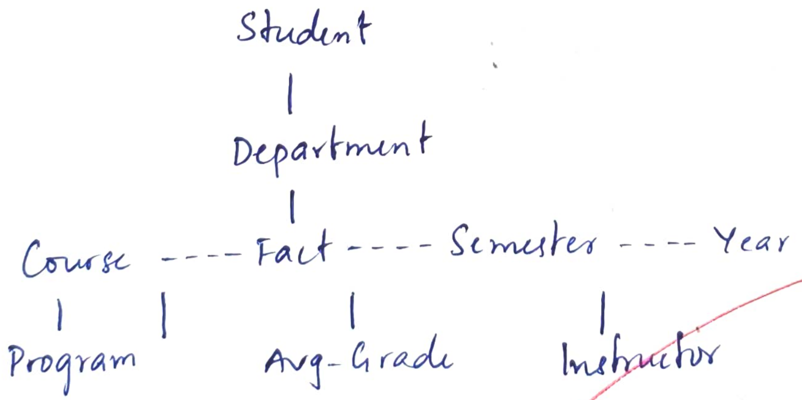
- Star schema
- Snowflake schema
- Galaxy schema

b, Star schema




c, SELECT doctor, patient, SUM(charge) AS total_charge
FROM fee
GROUP BY doctor, patient;

2, a,




b, OLAP operations

c, Number of cuboids

$= 5 \times 5 \times 5 \times 5 = 625$ cuboids

3, Star schema

Fact Table

Weather-Fact
- Air pressure
- Temperature
- Precipitation

Dimension tables

- Time (Hour → Day → Month → Year)
- Location (Probe → Region → Land/Ocean)
- Weather Type (Sunny, Rainy, Cloudy)

4, a, Maximum cells in base cuboid

$= p^n$

b, Minimum cells in base cuboid

$= p$

c, Maximum cells in entire cube

$= (p+1)^n$

d, Minimum cells in entire cube

$= 2^n$

5, a,

Student
|
Course ---- Fact-Academic ---- Faculty
|
Time

b, OLAP operations

1, Roll-up → Student to Department
2, Roll-up → Time to academic year
3, Aggregate → Calculate Average marks
4, Display department-wise average marks fro each year

c, ~~Total~~ ~~cuboids~~

⇒ $4 \times 3 \times 3 \times 5 = 180$ cuboids